

```
clear all
close all
s = tf('s')
```

s =

s

Continuous-time transfer function.
Model Properties

The system is

```
G = 1/(s*(s+1)*(0.5*s+1));
```

Spec 1: $K_v = 5$;

Spec 2: $\Phi_m > 40^\circ$

Spec 3: $g_m > 10\text{dB}$

Lag compensator

For a lag controller we have

$$C = K_c (T*s+1)/(\beta*T*s+1)$$

We need to design K_c , β , and T

Spec 1: $K_v = 20$

We compute K_v as

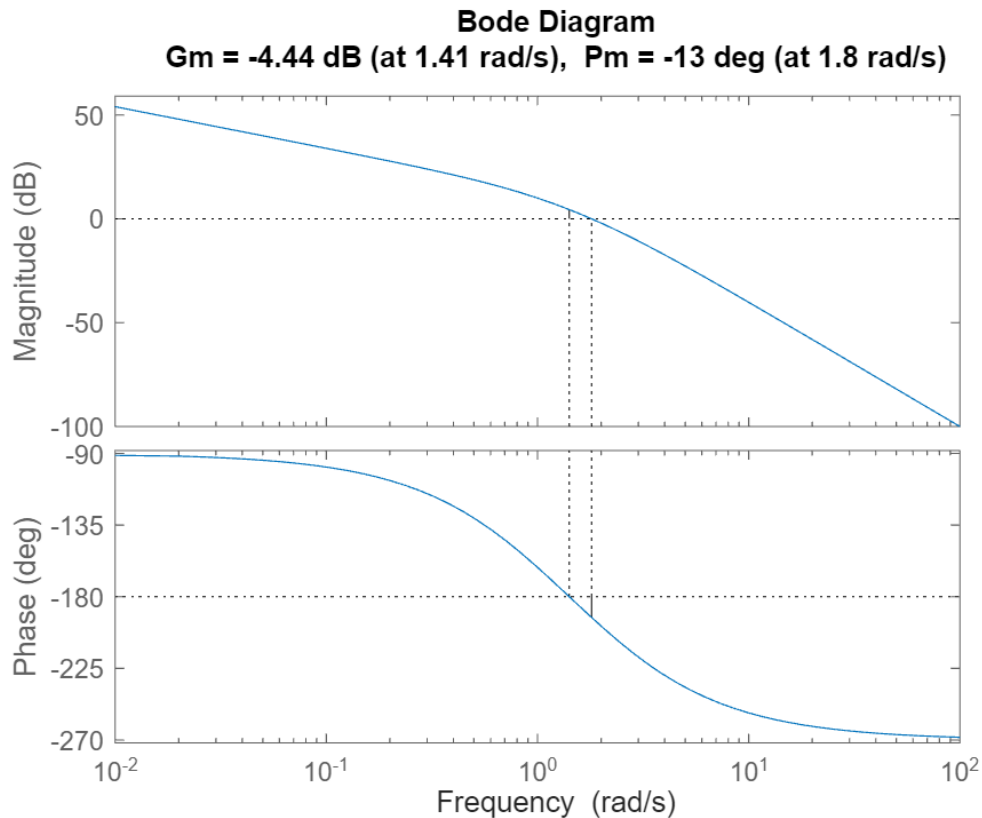
$$K_v = \lim_{s \rightarrow 0} s L(s) = K_c$$

Then, we choose

```
Kc = 5;
```

The bode plot with this gain is given by

```
margin(Kc*G)
```



Spec 2: $\phi_m > 40$

The goal is to achieve a phase margin of at least 40 degrees (targeting around 50). The desired gain crossover frequency is near 0.5 rad/s, where the magnitude is roughly 19 dB. We want to reduce the gain at this frequency to achieve the desired phase margin. To calculate beta, we use the relation:

$$20 \log (\beta) = 19 \text{ dB}$$

which gives

$$\beta = 10^{(19/20)};$$

Finally, to choose T, we place the zero of the lag compensator one decade below the new gain crossover frequency to ensure minimal impact on phase margin:

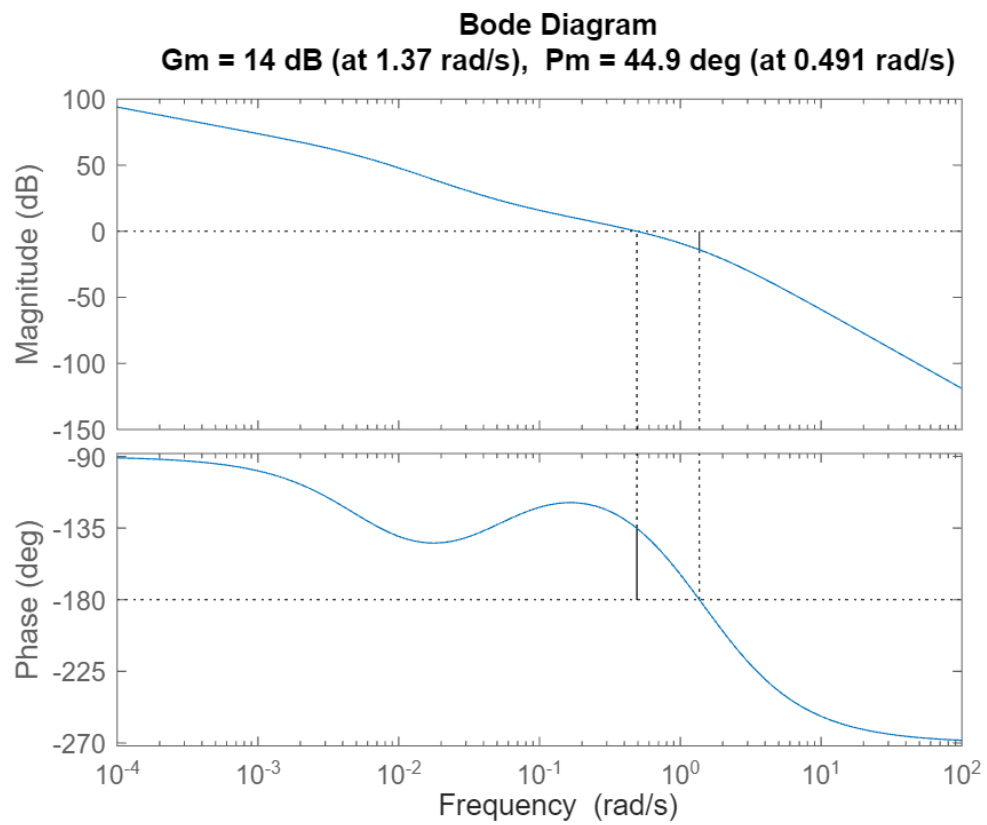
$$T = 10/0.5;$$

This gives the lag compensator

$$C = K_c * (T*s+1)/(\beta*T*s+1);$$

with bode plot

$$\text{margin}(C*G)$$



This design meets all specifications.